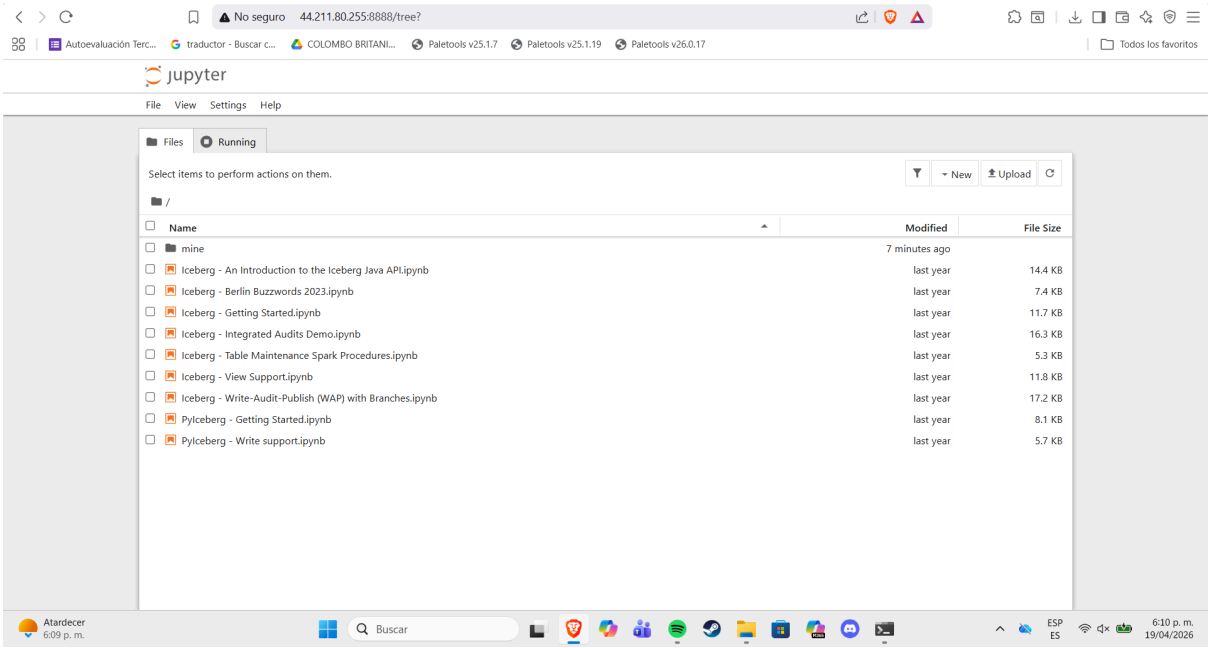
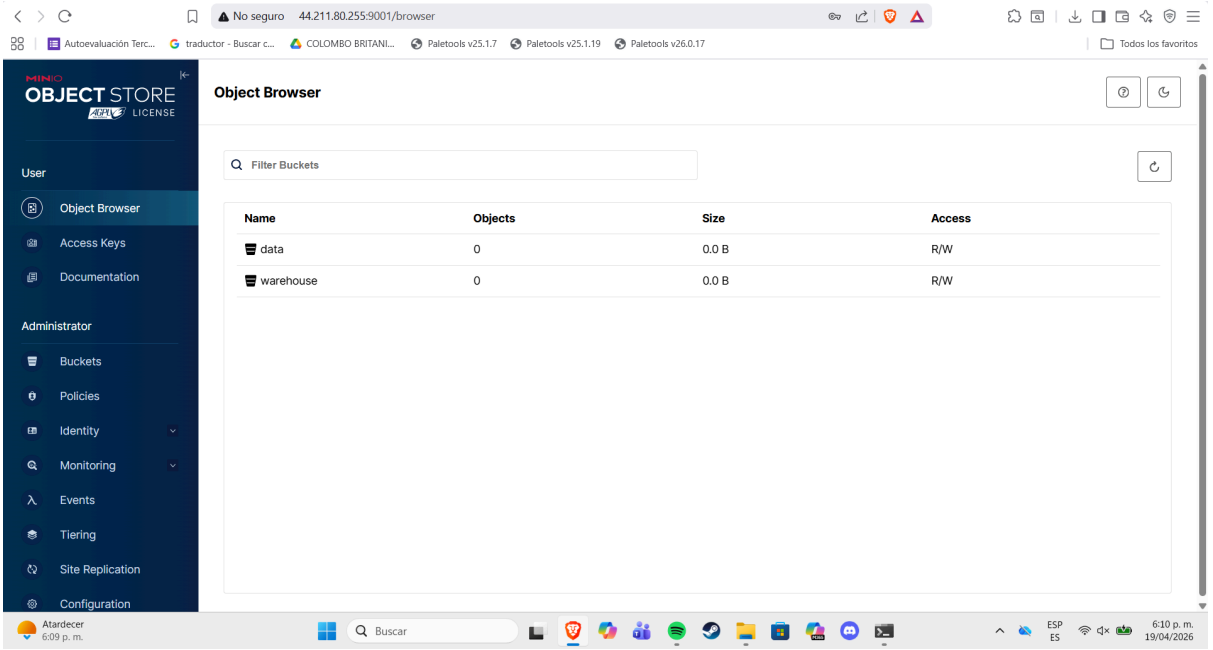
```
ubuntu@ip-172-31-86-152:~/Talleres-Sistemas-intensivos/Taller_5$ docker-compose ps
    Name                   Command                  State                          Ports
-----------------------------------------------------------------------------------------------------------
iceberg-rest    java -jar iceberg-rest-ima ...   Up              0.0.0.0:8181->8181/tcp,:::8181->8181/tcp
mc              /bin/sh -c                       Exit 0
                  sleep 5 &&
                  ...
minio           /usr/bin/docker-entrypoint ...   Up (unhealthy)  0.0.0.0:9000->9000/tcp,:::9000->9000/tcp,
                                                                 0.0.0.0:9001->9001/tcp,:::9001->9001/tcp
spark-iceberg   ./entrypoint.sh notebook         Up              0.0.0.0:10000->10000/tcp,:::10000->10000/tcp,
                                                                 0.0.0.0:10001->10001/tcp,:::10001->10001/tcp,
                                                                 0.0.0.0:8080->8080/tcp,:::8080->8080/tcp,
                                                                 0.0.0.0:8888->8888/tcp,:::8888->8888/tcp
```

SS2



SS3



SS3

jupyter taller5 Last Checkpoint: 11 minutes ago

File Edit View Run Kernel Settings Help — Not Trusted

Code — JupyterLab — Python 3 (ipykernel) — Format SQL

```
+-- metadata/     (snapshots, manifests, table metadata)
```

MinIO actua como object storage compatible con S3 — los datos viven como archivos Parquet pero Iceberg agrega una capa de metadatos que habilita las capacidades ACID.

[1]:
```python
from pyspark.sql import SparkSession
from pyspark.sql import functions as F

spark = SparkSession.builder \
    .appName("Taller5-Iceberg") \
    .config("spark.sql.extensions", "org.apache.iceberg.spark.extensions.IcebergSparkSessionExtensions") \
    .config("spark.sql.catalog.demo", "org.apache.iceberg.spark.SparkCatalog") \
    .config("spark.sql.catalog.demo.type", "rest") \
    .config("spark.sql.catalog.demo.uri", "http://rest:8181") \
    .config("spark.sql.catalog.demo.io-impl", "org.apache.iceberg.aws.s3.S3FileIO") \
    .config("spark.sql.catalog.demo.warehouse", "s3://warehouse/") \
    .config("spark.sql.catalog.demo.s3.endpoint", "http://minio:9000") \
    .config("spark.sql.defaultCatalog", "demo") \
    .getOrCreate()

spark.sparkContext.setLogLevel("WARN")
print("SparkSession creada con catalogo Iceberg REST")
print(f"Version de Spark: {spark.version}")
```

```
SparkSession creada con catalogo Iceberg REST
Version de Spark: 3.5.5
26/04/19 23:12:42 WARN SparkSession: Using an existing Spark session; only runtime SQL configurations will take effect.
```

SS4

No seguro 44.211.80.255:8888/notebooks/mine/taller5.ipynb

Autoevaluación Terc... traductor - Buscar c... COLOMBO BRITANI... Paletools v25.1.7 Paletools v25.1.19 Paletools v26.0.17 Todos los favoritos

jupyter taller5 Last Checkpoint: 30 minutes ago

File Edit View Run Kernel Settings Help — Trusted

Code — JupyterLab — Python 3 (ipykernel) — Format SQL

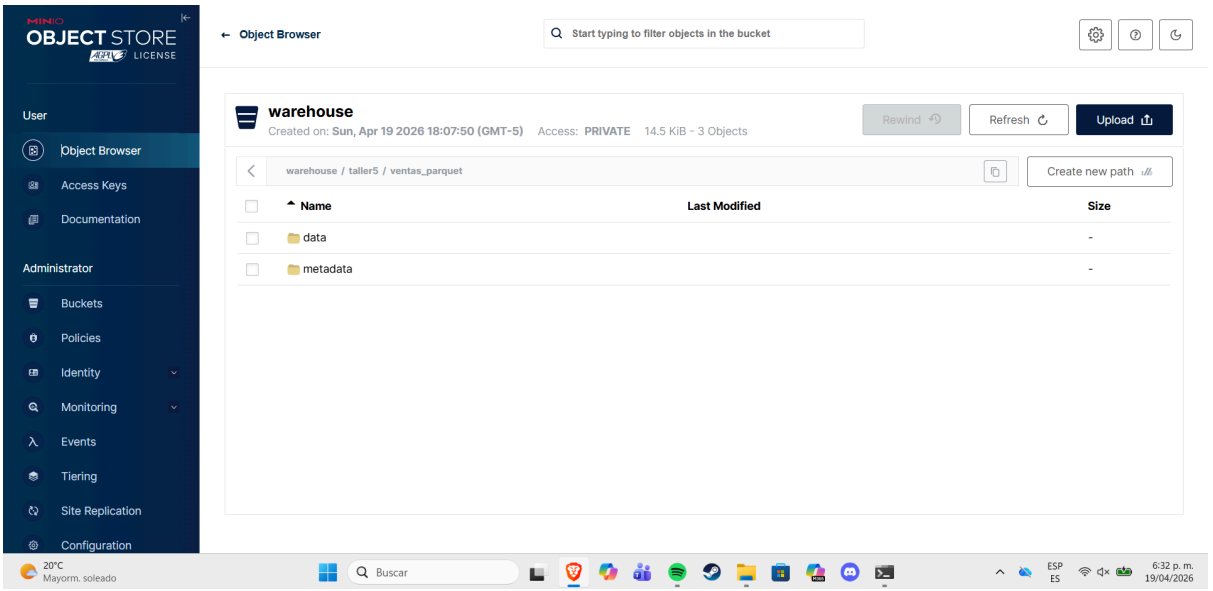
```python
# Escribir como tabla temporal Iceberg para simular Parquet plano
spark.sql("DROP TABLE IF EXISTS demo.taller5.ventas_parquet")
df.writeTo("demo.taller5.ventas_parquet").create()
print("\nEscrito como tabla en demo.taller5.ventas_parquet")
print("PROBLEMA con Parquet plano: no tiene ACID, no tiene historial, no tiene time travel")
```

```
Registros cargados: 50
root
 |-- id: integer (nullable = true)
 |-- fecha: date (nullable = true)
 |-- producto: string (nullable = true)
 |-- categoria: string (nullable = true)
 |-- cantidad: integer (nullable = true)
 |-- precio_unitario: integer (nullable = true)
 |-- region: string (nullable = true)

+---+----------+----------+----------+--------+---------------+------------+
| id|     fecha|  producto| categoria|cantidad|precio_unitario|      region|
+---+----------+----------+----------+--------+---------------+------------+
|  1|2024-01-03|    Laptop|Electronica|       1|        1500000|      Bogota|
|  2|2024-01-05|   Teclado|Perifericos|       3|          85000|    Medellin|
|  3|2024-01-08|   Monitor|Electronica|       2|         750000|        Cali|
|  4|2024-01-10|     Mouse|Perifericos|       5|          45000|Barranquilla|
|  5|2024-01-12|Auriculares|Perifericos|       2|         180000| Bucaramanga|
+---+----------+----------+----------+--------+---------------+------------+
only showing top 5 rows


Escrito como tabla en demo.taller5.ventas_parquet
PROBLEMA con Parquet plano: no tiene ACID, no tiene historial, no tiene time travel
```

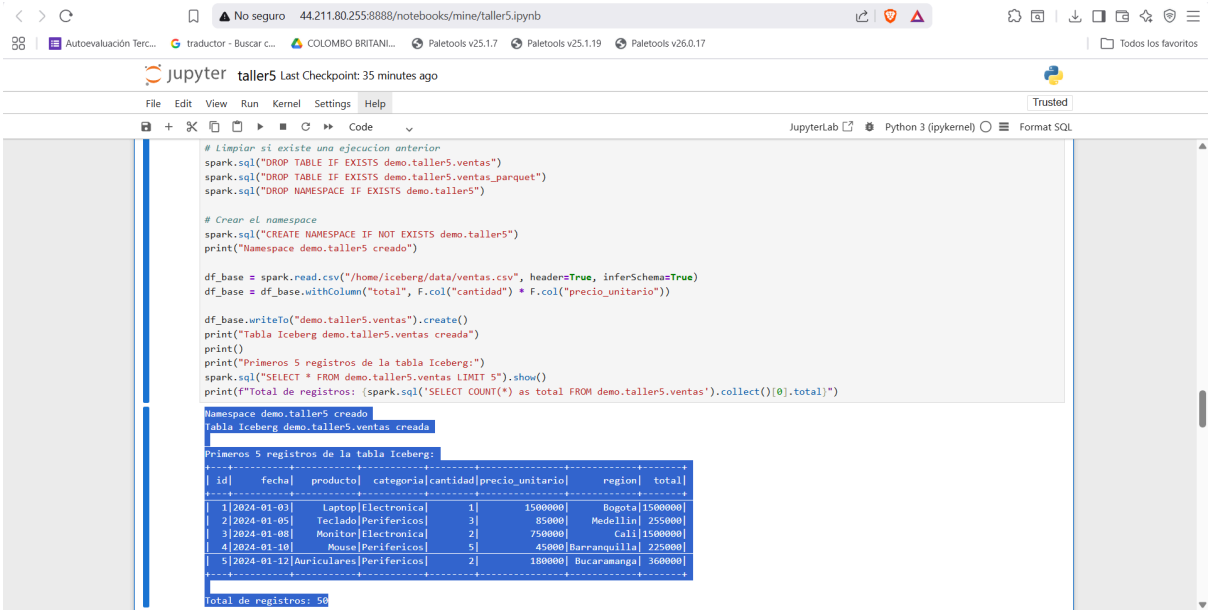
[ ]:
```python
# Intentar un UPDATE sobre Parquet plano — esto no existe en la API de Spark DataFrames
```

20°C Mayorm. soleado — Buscar — ESP ES — 6:31 p. m. 19/04/2026

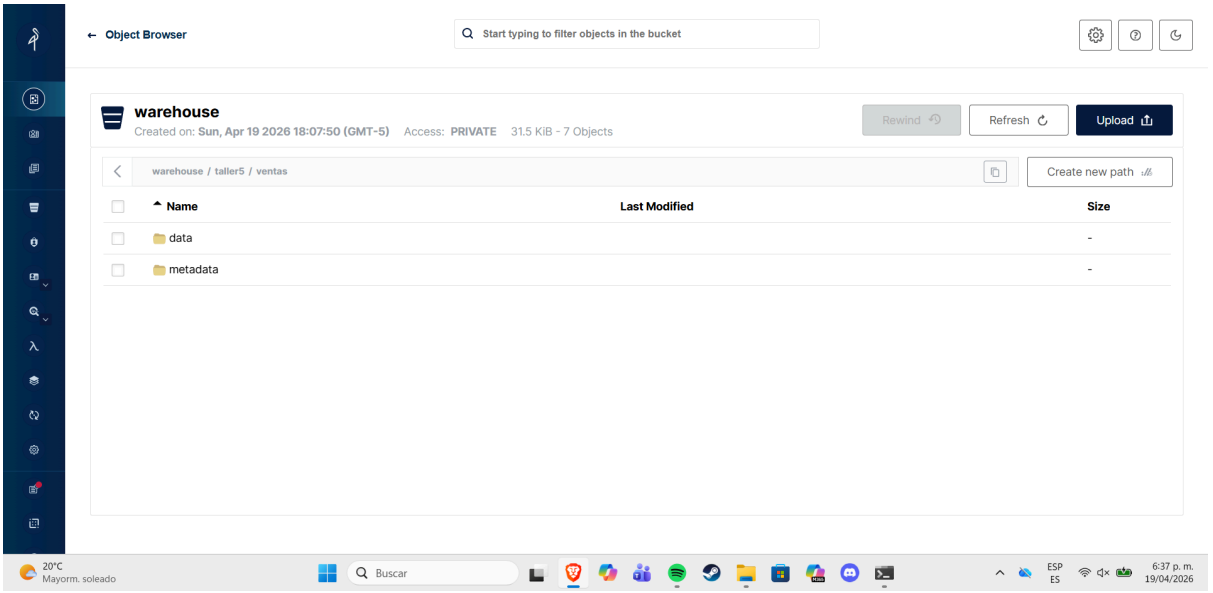
SS5


MINIO
OBJECT STORE
AGPLv3 LICENSE
User
Object Browser
Access Keys
Documentation
Administrator
Buckets
Policies
Identity
Monitoring
Events
Tiering
Site Replication
Configuration
Object Browser
Start typing to filter objects in the bucket
warehouse
Created on: Sun, Apr 19 2026 18:07:50 (GMT-5)
Access: PRIVATE
14.5 KiB - 3 Objects
Rewind
Refresh
Upload
warehouse / taller5 / ventas_parquet
Create new path
Name
Last Modified
Size
data
-
metadata
-
20°C
Mayorm. soleado
Buscar
ESP
ES
6:32 p. m.
19/04/2026

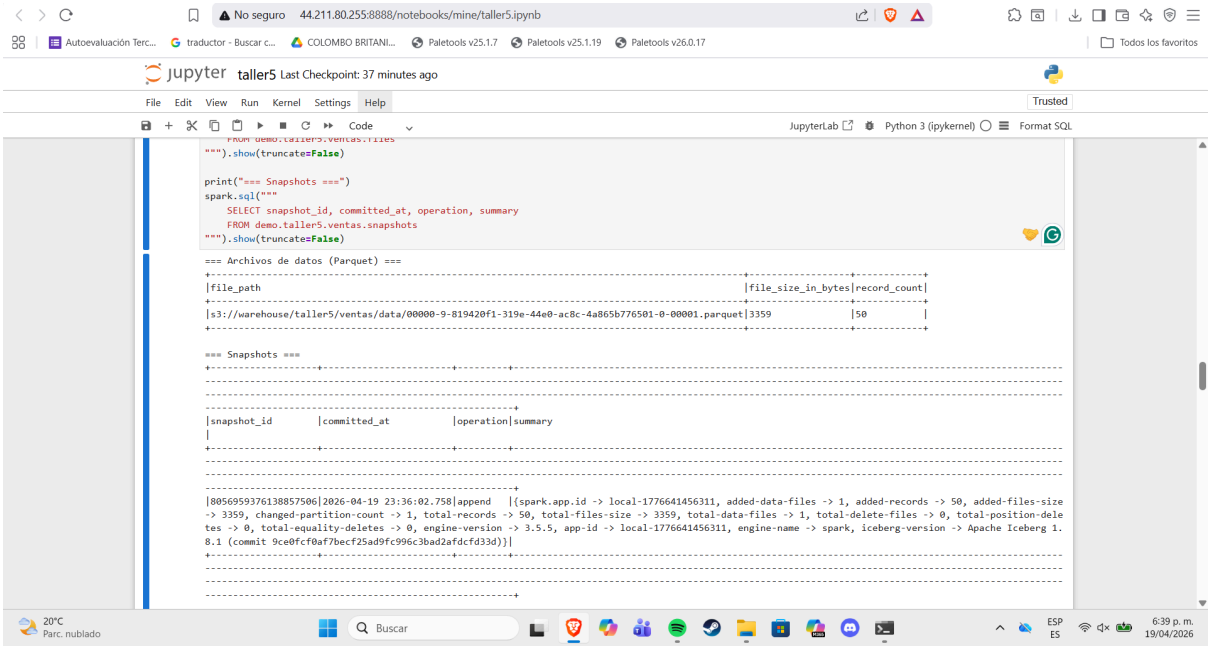
SS6


No seguro
44.211.80.255:8888/notebooks/mine/taller5.ipynb
Autoevaluación Terc...
traductor - Buscar c...
COLOMBO BRITANI...
Paletools v25.1.7
Paletools v25.1.19
Paletools v26.0.17
Todos los favoritos
jupyter
taller5 Last Checkpoint: 35 minutes ago
File
Edit
View
Run
Kernel
Settings
Help
Trusted
Code
JupyterLab
Python 3 (ipykernel)
Format SQL
# Limpiar si existe una ejecucion anterior
spark.sql("DROP TABLE IF EXISTS demo.taller5.ventas")
spark.sql("DROP TABLE IF EXISTS demo.taller5.ventas_parquet")
spark.sql("DROP NAMESPACE IF EXISTS demo.taller5")
# Crear el namespace
spark.sql("CREATE NAMESPACE IF NOT EXISTS demo.taller5")
print("Namespace demo.taller5 creado")
df_base = spark.read.csv("/home/iceberg/data/ventas.csv", header=True, inferSchema=True)
df_base = df_base.withColumn("total", F.col("cantidad") * F.col("precio_unitario"))
df_base.writeTo("demo.taller5.ventas").create()
print("Tabla Iceberg demo.taller5.ventas creada")
print()
print("Primeros 5 registros de la tabla Iceberg:")
spark.sql("SELECT * FROM demo.taller5.ventas LIMIT 5").show()
print(f"Total de registros: {spark.sql('SELECT COUNT(*) as total FROM demo.taller5.ventas').collect()[0].total}")
Namespace demo.taller5 creado
Tabla Iceberg demo.taller5.ventas creada
Primeros 5 registros de la tabla Iceberg:
+---+----------+----------+----------+--------+---------------+----------+-------+
| id| fecha| producto| categoria|cantidad|precio_unitario| region| total|
+---+----------+----------+----------+--------+---------------+----------+-------+
1	2024-01-03	Laptop	Electronica	1	1500000	Bogota	1500000
2	2024-01-05	Teclado	Perifericos	3	85000	Medellin	255000
3	2024-01-08	Monitor	Electronica	2	750000	Cali	1500000
4	2024-01-10	Mouse	Perifericos	5	45000	Barranquilla	225000
5	2024-01-12	Auriculares	Perifericos	2	180000	Bucaramanga	360000
+---+----------+----------+----------+--------+---------------+----------+-------+
Total de registros: 50

SS7


Object Browser
Start typing to filter objects in the bucket
warehouse
Created on: Sun, Apr 19 2026 18:07:50 (GMT-5)
Access: PRIVATE
31.5 KiB - 7 Objects
Rewind
Refresh
Upload
warehouse / taller5 / ventas
Create new path
Name
Last Modified
Size
data
-
metadata
-
20°C
Mayorm. soleado
Buscar
ESP
ES
6:37 p. m.
19/04/2026

SS8


No seguro
44.211.80.255:8888/notebooks/mine/taller5.ipynb
jupyter
taller5 Last Checkpoint: 37 minutes ago
File
Edit
View
Run
Kernel
Settings
Help
Trusted
Code
JupyterLab
Python 3 (ipykernel)
Format SQL
""").show(truncate=False)
print("=== Snapshots ===")
spark.sql("""
SELECT snapshot_id, committed_at, operation, summary
FROM demo.taller5.ventas.snapshots
""").show(truncate=False)
=== Archivos de datos (Parquet) ===
+----------+----------+
|file_path|file_size_in_bytes|record_count|
+----------+----------+----------+
|s3://warehouse/taller5/ventas/data/00000-9-819420f1-319e-44e0-ac8c-4a865b776501-0-00001.parquet|3359|50|
+----------+----------+----------+
=== Snapshots ===
+----------+----------+---------+----------+
|snapshot_id|committed_at|operation|summary|
+----------+----------+---------+----------+
|8056959376138857506|2026-04-19 23:36:02.758|append|{spark.app.id -> local-1776641456311, added-data-files -> 1, added-records -> 50, added-files-size -> 3359, changed-partition-count -> 1, total-records -> 50, total-files-size -> 3359, total-data-files -> 1, total-delete-files -> 0, total-position-deletes -> 0, total-equality-deletes -> 0, engine-version -> 3.5.5, app-id -> local-1776641456311, engine-name -> spark, iceberg-version -> Apache Iceberg 1.8.1 (commit 9ce0fcf0af7becf25ad9fc996c3bad2afdcfd33d)}|
+----------+----------+---------+----------+
20°C
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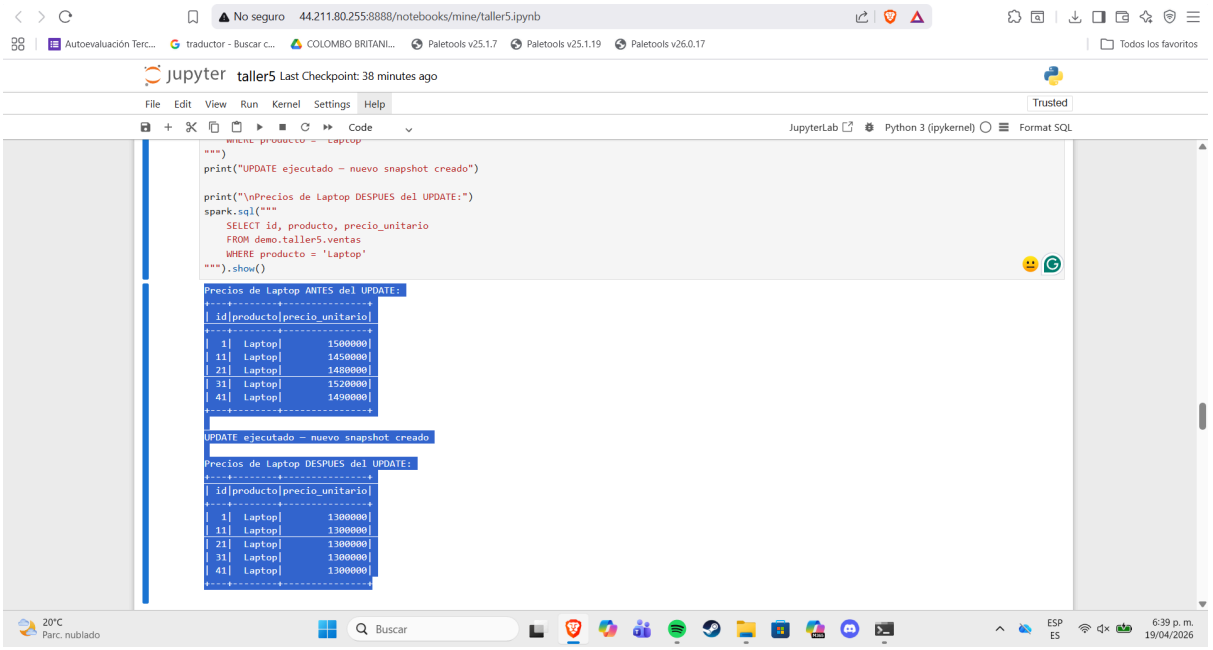
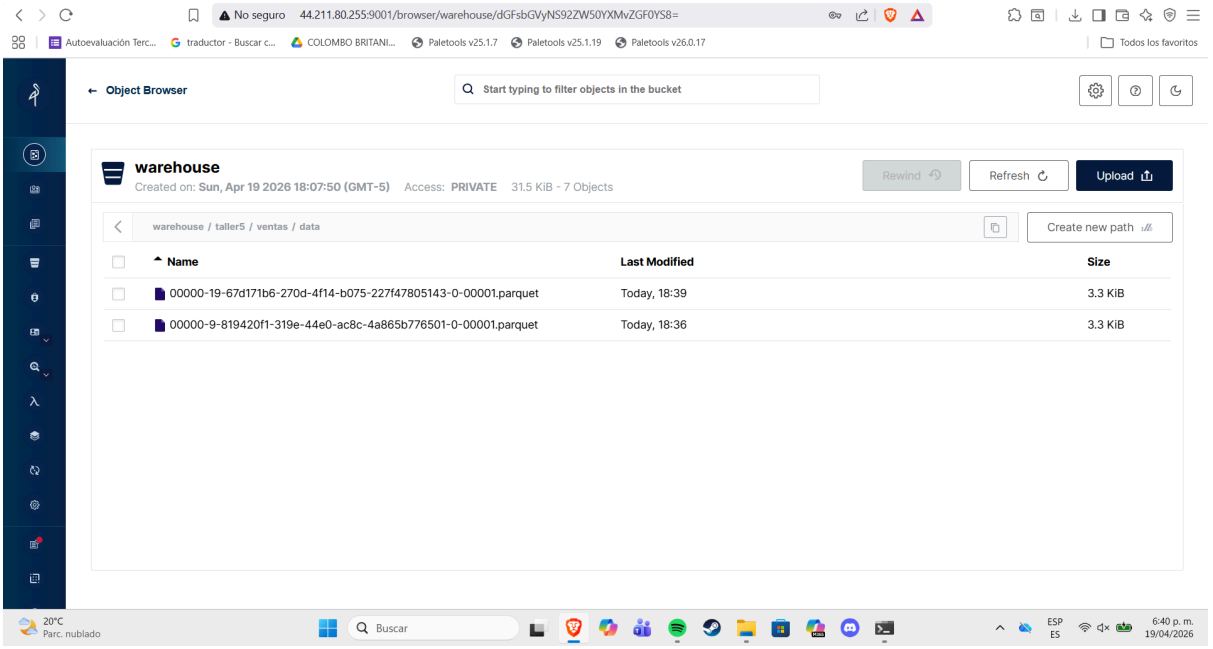
Buscar
ESP
ES
6:39 p. m.
19/04/2026

SS9


No seguro 44.211.80.255:8888/notebooks/mine/taller5.ipynb
jupyter taller5 Last Checkpoint: 38 minutes ago
File Edit View Run Kernel Settings Help
Trusted
Code
JupyterLab
Python 3 (ipykernel)
Format SQL
"""
print("UPDATE ejecutado — nuevo snapshot creado")
print("\nPrecios de Laptop DESPUES del UPDATE:")
spark.sql("""
SELECT id, producto, precio_unitario
FROM demo.taller5.ventas
WHERE producto = 'Laptop'
""").show()
Precios de Laptop ANTES del UPDATE:
+---+--------+---------------+
| id|producto|precio_unitario|
+---+--------+---------------+
1	Laptop	1500000
11	Laptop	1450000
21	Laptop	1480000
31	Laptop	1520000
41	Laptop	1490000
+---+--------+---------------+
UPDATE ejecutado — nuevo snapshot creado
Precios de Laptop DESPUES del UPDATE:
+---+--------+---------------+
| id|producto|precio_unitario|
+---+--------+---------------+
1	Laptop	1300000
11	Laptop	1300000
21	Laptop	1300000
31	Laptop	1300000
41	Laptop	1300000
+---+--------+---------------+

SS10


No seguro 44.211.80.255:9001/browser/warehouse/dGFsbGVyNS92ZW50YXMvZGF0YS8=
Object Browser
Start typing to filter objects in the bucket
warehouse
Created on: Sun, Apr 19 2026 18:07:50 (GMT-5) Access: PRIVATE 31.5 KiB - 7 Objects
Rewind
Refresh
Upload
warehouse / taller5 / ventas / data
Create new path
Name
Last Modified
Size
00000-19-67d171b6-270d-4f14-b075-227f47805143-0-00001.parquet
Today, 18:39
3.3 KiB
00000-9-819420f1-319e-44e0-ac8c-4a865b776501-0-00001.parquet
Today, 18:36
3.3 KiB

SS11

No seguro 44.211.80.255:8888/notebooks/mine/taller5.ipynb

jupyter taller5 Last Checkpoint: 40 minutes ago

File Edit View Run Kernel Settings Help

```
spark.sql("SELECT COUNT(*) as total FROM demo.taller5.ventas").show()

print("Snapshots acumulados hasta ahora:")
spark.sql("""
    SELECT snapshot_id, made_current_at, is_current_ancestor
    FROM demo.taller5.ventas.history
""").show(truncate=False)
```

```
Total de registros ANTES del DELETE:
+-----+
|total|
+-----+
|   50|
+-----+

Registros que cumplen la condicion de DELETE: 0
DELETE ejecutado — nuevo snapshot creado

Total de registros DESPUES del DELETE:
+-----+
|total|
+-----+
|   50|
+-----+

Snapshots acumulados hasta ahora:
+-------------------+-----------------------+-------------------+
|snapshot_id        |made_current_at        |is_current_ancestor|
+-------------------+-----------------------+-------------------+
|8056959376138857506|2026-04-19 23:36:02.758|true               |
|7929112922914506546|2026-04-19 23:39:30.322|true               |
|2945640966659751080|2026-04-19 23:40:58.438|true               |
|2809549611853507490|2026-04-19 23:41:39.995|true               |
+-------------------+-----------------------+-------------------+
```

SS12

No seguro 44.211.80.255:8888/notebooks/mine/taller5.ipynb

jupyter taller5 Last Checkpoint: 43 minutes ago

File Edit View Run Kernel Settings Help

```
""").show()

print("=== Precios de Laptop en el snapshot ACTUAL ===")
spark.sql("""
    SELECT id, producto, precio_unitario
    FROM demo.taller5.ventas
    WHERE producto = 'Laptop'
""").show()
```

```
Primer snapshot (INSERT inicial): 8056959376138857506
Snapshot actual (mas reciente):   2809549611853507490

=== TIME TRAVEL: Precios de Laptop en el snapshot INICIAL ===
+---+--------+---------------+
| id|producto|precio_unitario|
+---+--------+---------------+
|  1|  Laptop|        1500000|
| 11|  Laptop|        1450000|
| 21|  Laptop|        1480000|
| 31|  Laptop|        1520000|
| 41|  Laptop|        1490000|
+---+--------+---------------+

=== Precios de Laptop en el snapshot ACTUAL ===
+---+--------+---------------+
| id|producto|precio_unitario|
+---+--------+---------------+
|  1|  Laptop|        1300000|
| 11|  Laptop|        1300000|
| 21|  Laptop|        1300000|
| 31|  Laptop|        1300000|
| 41|  Laptop|        1300000|
+---+--------+---------------+
```

SS13

No seguro 44.211.80.255:8888/notebooks/mine/taller5.ipynb

jupyter taller5 Last Checkpoint: 44 minutes ago

File Edit View Run Kernel Settings Help

Trusted

Code

JupyterLab Python 3 (ipykernel) Format SQL

```
    SELECT id, producto, precio_unitario, descuento
    FROM demo.taller5.ventas
    LIMIT 5
""").show()

print("La columna nueva aparece como NULL en filas anteriores — sin reescribir datos")
```

```
Schema ANTES de agregar columna:
+---------------+---------+-------+
|       col_name|data_type|comment|
+---------------+---------+-------+
|             id|      int|   NULL|
|          fecha|     date|   NULL|
|       producto|   string|   NULL|
|      categoria|   string|   NULL|
|       cantidad|      int|   NULL|
|precio_unitario|      int|   NULL|
|         region|   string|   NULL|
|          total|      int|   NULL|
+---------------+---------+-------+

Columna 'descuento' agregada al schema

Filas existentes con la nueva columna (muestra NULL — sin reescritura):
+---+----------+---------------+---------+
| id|  producto|precio_unitario|descuento|
+---+----------+---------------+---------+
|  1|    Laptop|        1300000|     NULL|
|  2|   Teclado|          85000|     NULL|
|  3|   Monitor|         750000|     NULL|
|  4|     Mouse|          45000|     NULL|
|  5|Auriculares|         180000|     NULL|
+---+----------+---------------+---------+

La columna nueva aparece como NULL en filas anteriores — sin reescribir datos
```

20°C Parc. nublado Buscar ESP ES 6:46 p. m. 19/04/2026

SS14

No seguro 44.211.80.255:8888/notebooks/mine/taller5.ipynb

jupyter taller5 Last Checkpoint: 45 minutes ago

File Edit View Run Kernel Settings Help

Trusted

Code

JupyterLab Python 3 (ipykernel) Format SQL

```
nuevas_filas = spark.createDataFrame([
    (51, date(2024, 6, 1), "Laptop", "Electronica", 2, 1300000, "Bogota", 2600000, 0.10),
    (52, date(2024, 6, 5), "Smartphone", "Movil", 1, 1200000, "Medellin", 1200000, 0.05),
], ["id", "fecha", "producto", "categoria", "cantidad", "precio_unitario", "region", "total", "descuento"])

nuevas_filas.writeTo("demo.taller5.ventas").append()
print("Nuevas filas con descuento insertadas")

print("\nMix de filas antiguas (descuento=NULL) y nuevas (descuento poblado):")
spark.sql("""
    SELECT id, producto, precio_unitario, descuento
    FROM demo.taller5.ventas
    WHERE id >= 49
    ORDER BY id
""").show()

print(f"Total de registros finales: {spark.sql('SELECT COUNT(*) as total FROM demo.taller5.ventas').collect()[0].total}")
```

```
Nuevas filas con descuento insertadas

Mix de filas antiguas (descuento=NULL) y nuevas (descuento poblado):
+---+----------+---------------+---------+
| id|  producto|precio_unitario|descuento|
+---+----------+---------------+---------+
| 49|Smartphone|        1320000|     NULL|
| 50|    Webcam|         230000|     NULL|
| 51|    Laptop|        1300000|      0.1|
| 52|Smartphone|        1200000|     0.05|
+---+----------+---------------+---------+

Total de registros finales: 52
```

20°C Parc. nublado Buscar ESP ES 6:46 p. m. 19/04/2026